MSc Data Science
DS601: Mathematical Foundations for Data Science
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Tutorial 2

1. In a laboratory probability experiment, a special regular tetrahedron (a 4-sided die) is rolled. The four faces are colored as follows:

   - Face 1 is Solid Red.
   - Face 2 is Solid Blue.
   - Face 3 is Solid Green.
   - Face 4 is a mix of all three colors (Red, Blue, and Green).

   Let R, B, and G be the events that the face landing face-down contains the color Red, Blue, and Green, respectively.

   a. Prove whether the events R, B, and G are pairwise independent.
   b. Determine whether the events are mutually independent.

2. Let A and B be two independent events in a sample space. You are given the following information:

   - The probability that at least one of the events occurs is 0.88.
   - The probability that exactly one of the events occurs is 0.56.

   Find the individual probabilities $P(A)$ and $P(B)$.

3. A jar contains 6 red marbles and 4 blue marbles. Two marbles are drawn from the jar sequentially without replacement. What is the probability that the second marble drawn is red, given that the first marble drawn was blue? (Class Activity)

4. A digital signal consisting of zeros and ones is transmitted through a network with one relay station in series (Source $\rightarrow$ Relay 1 $\rightarrow$ Receiver). The source transmits a '1' with a probability of 0.6 and a '0' with a probability of 0.4.

   During each leg of the transmission, noise in the channel flips the bit (changes a 0 to a 1, or a 1 to a 0) with a probability of $p = 0.1$. The one transmission leg operates completely independently of one another.

   If the final receiver observes a '1', what is the conditional probability that the source originally transmitted a '1'?

5. Three factories F_1, F_2 and F_3 produce 20%, 30%, and 50% of the products, respectively. Their probabilities of producing a defective item are 1%, 2%, and 4%, respectively. Two products are selected independently. Find the probability that exactly one of the two products is defective.

6. A student travels to college using one of three modes:
   Bus: 50% of the trips
   Auto: 30% of the trips
   Bike: 20% of the trips

The probability that the student is late depends on the mode of transport:
$P(L|\text{Bus}) = 0.10 \quad P(L|\text{Auto}) = 0.20 \quad P(L|\text{Bike}) = 0.05$
However, if it rains, the probability of being late for each mode changes to:
Bus: 0.30 Auto: 0.40 Bike: 0.15
The probability that it rains on a randomly chosen day is 30%. Assume that the mode of transport chosen by the student is independent of whether it rains. Find the probability that the student is late on a randomly chosen trip.

7. A box contains 6 Red, 5 Blue, and 4 Green balls. Two balls are drawn successively without replacement. Answer the following questions:

   (1) Determine the probability that both balls drawn are Red.

   (2) Given that the first ball drawn is Red, determine the probability that the second ball drawn is Blue.

   (3) After the two balls have been drawn, one of the two balls is selected uniformly at random and is observed to be Blue. What is the probability that this Blue ball was the first ball drawn?

   (4) Determine the probability that the two balls drawn are of different colors.

8. A player rolls two fair dice in a game. The rules are:

   - If the sum is 7 or 11, the player wins.
   - If both dice show the same number, the player receives a bonus.
   - If the sum is 3 or less, the player loses immediately.

   If the two dice are rolled randomly, find:

   (a) The probability that the player wins.

   (b) The probability that the player receives a bonus or loses immediately.

   (c) Can we simply add the probabilities of winning, receiving a bonus, and losing immediately to get 1? Give a reason for your answer.